

The arm

Department of Anatomy

Musculoskeletal System

The Upper Limb (5)

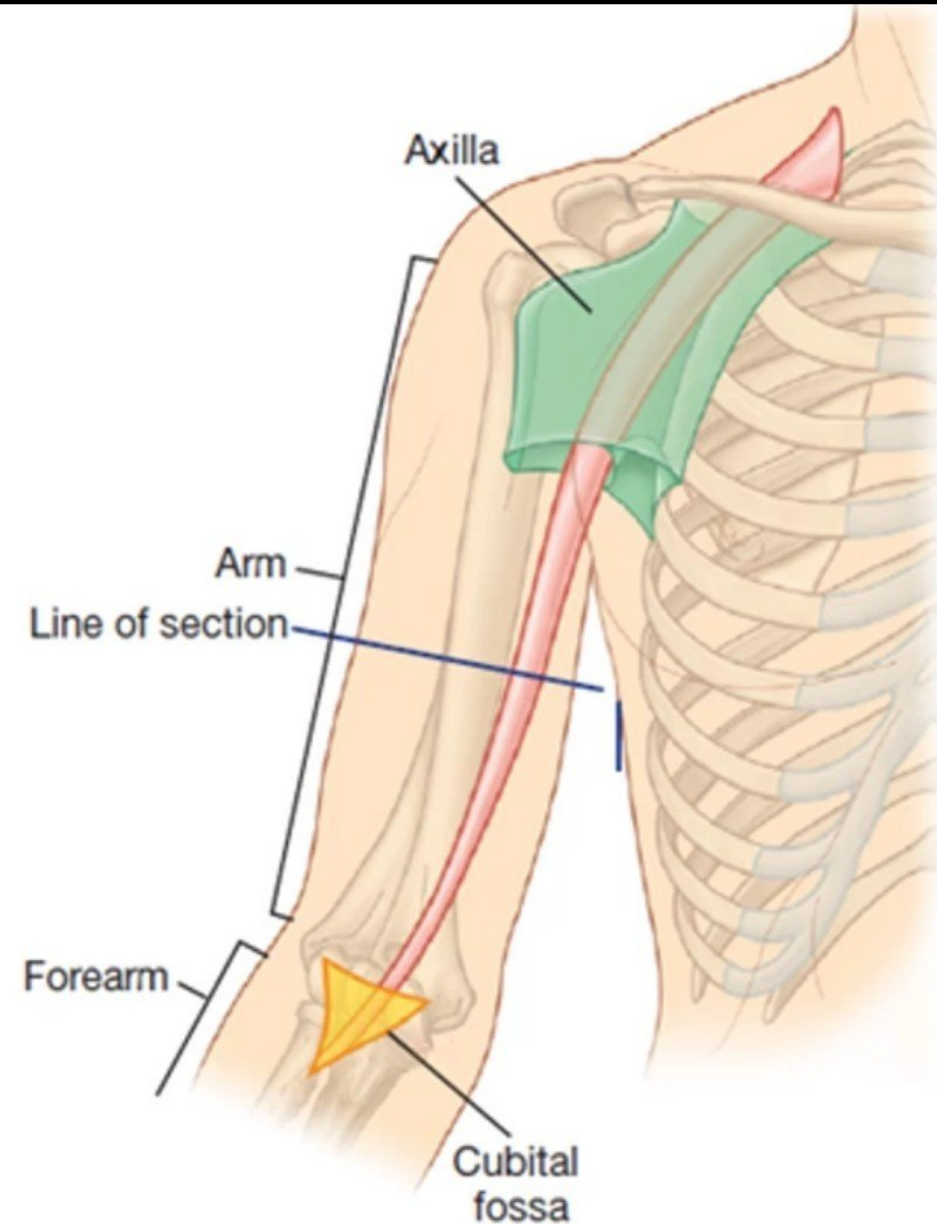
Dr. Osman Ahmed Osman

Lecture points

- **Muscles of the arm**
- **Nerves of the arm**
- **Blood supply**
- **Cubital fossa**
- **Clinical notes**

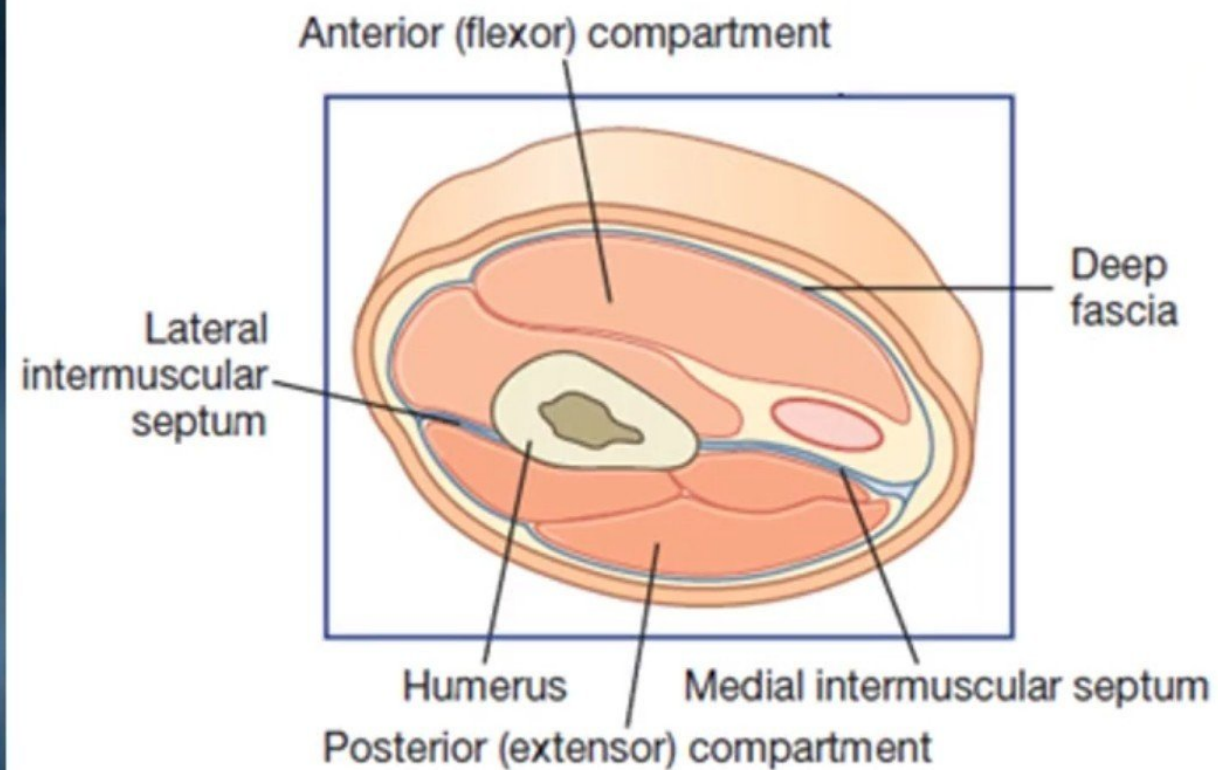
The arm

- It is region of the upper limb between **the shoulder** and **the elbow** .
- The **superior aspect** of the arm communicates **medially** with **the axilla**.
- **Inferiorly**, the cubital fossa is positioned **anterior** to **the elbow joint**.



The arm is divided into two compartments by medial and lateral intramuscular septa into :

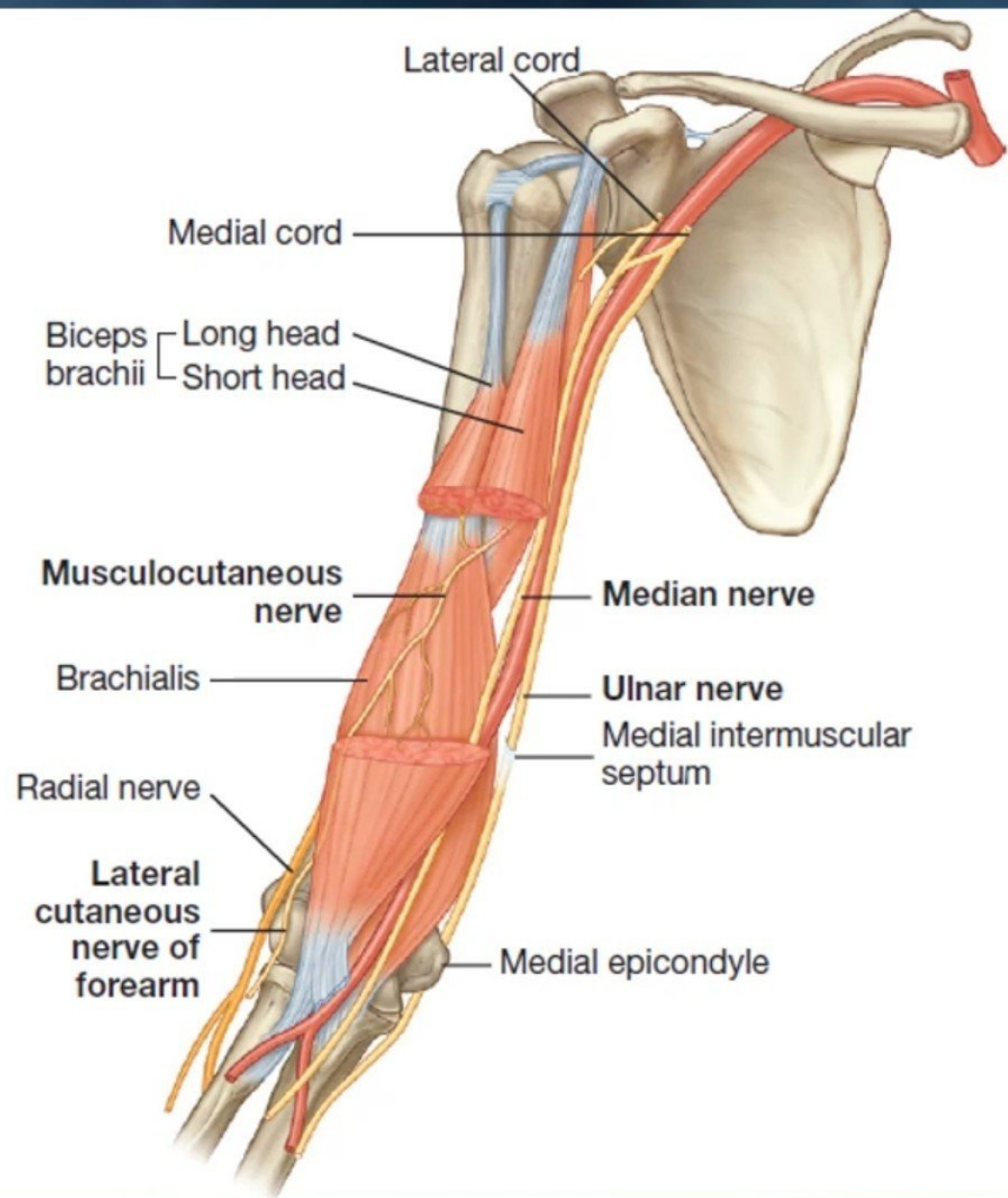
- **Anterior compartment of the arm** contains muscles that predominantly **flex the elbow joint**.
- **Posterior compartment of the arm** contains muscles that **extend the elbow joint**.

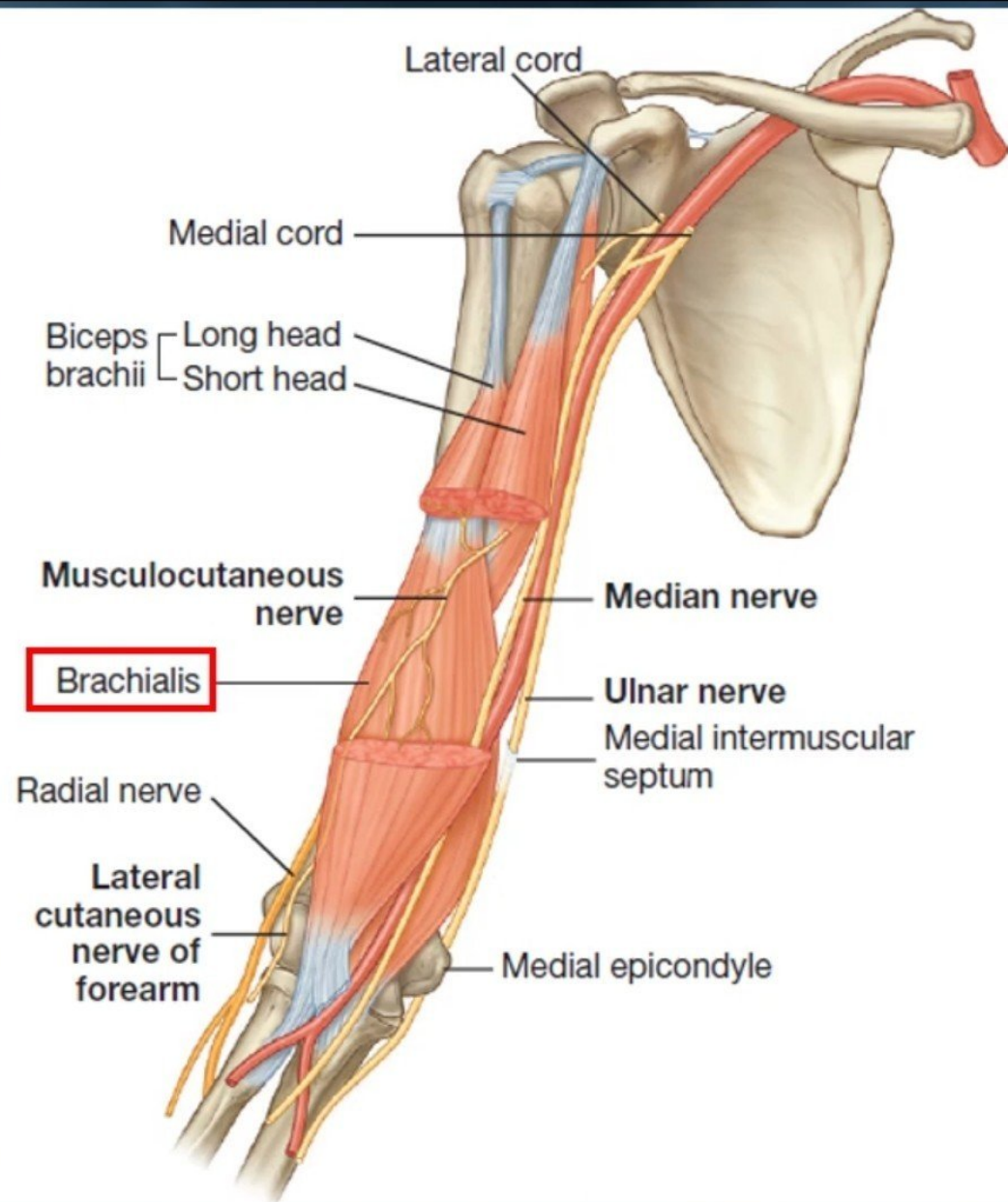
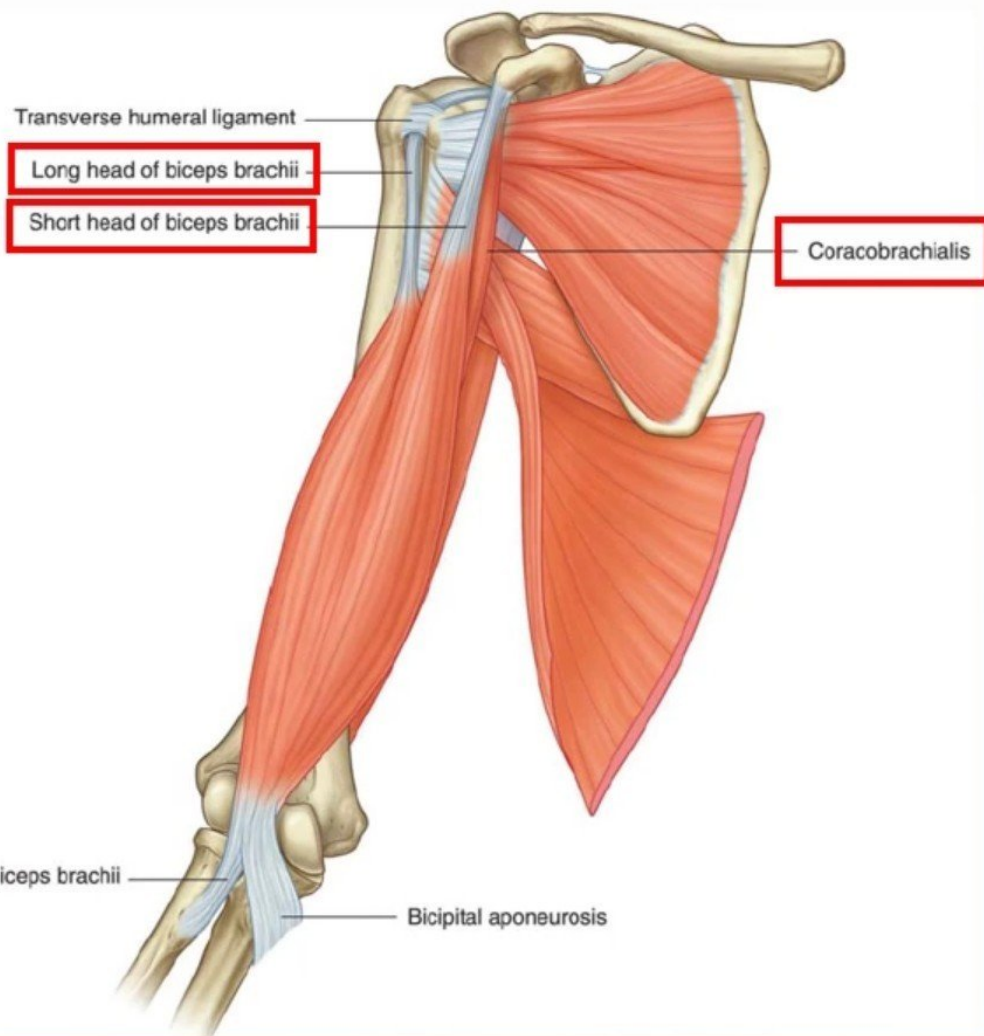


- The anterior compartment of the arm contains three muscles

1. Coracobrachialis
2. Brachialis
3. Biceps brachii

- They are innervated by the **musculocutaneous nerve**.





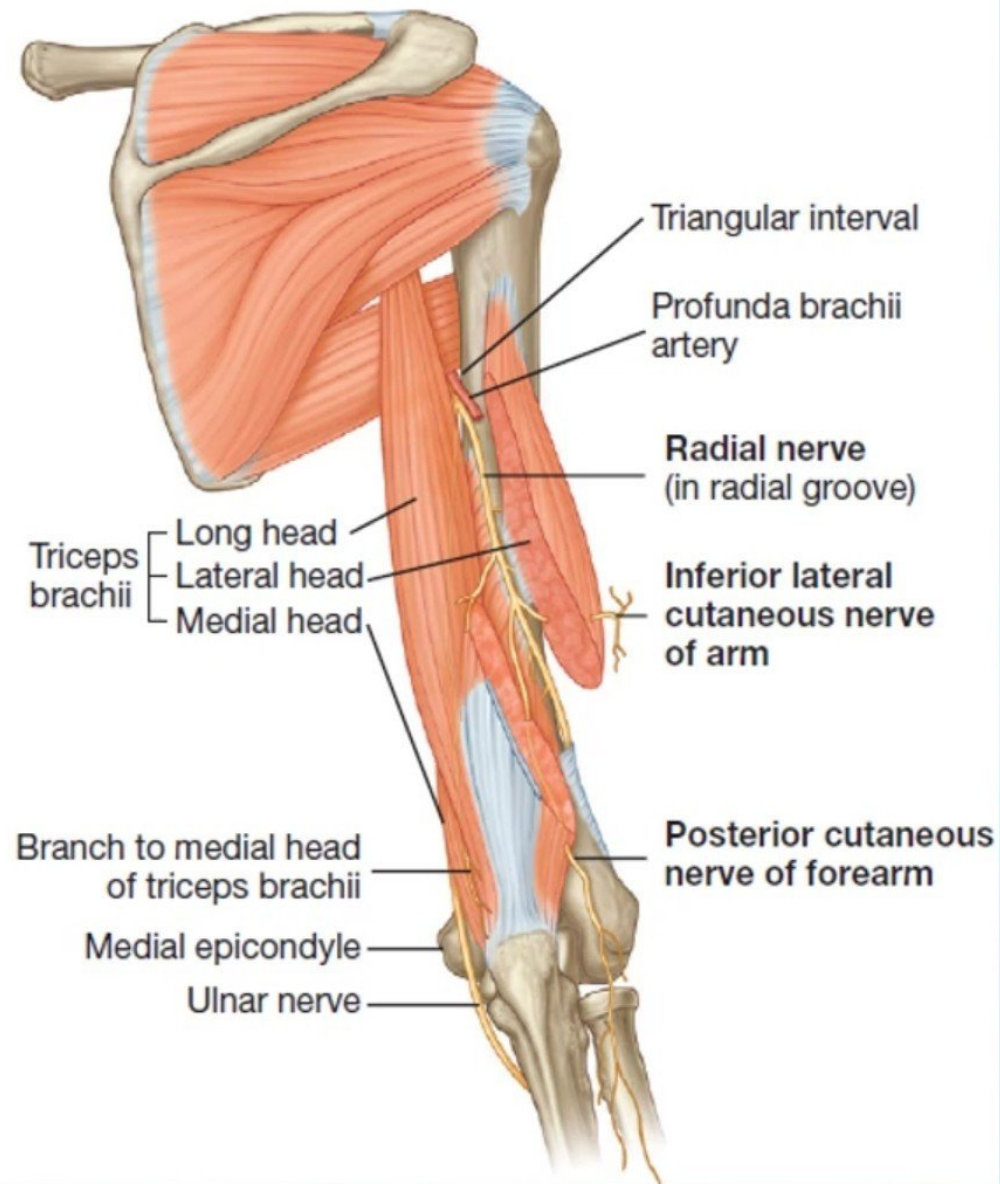
The muscles of the anterior compartment of the arm:

Muscles of the anterior compartment of the arm (spinal segments indicated in **bold** are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Coracobrachialis	Apex of coracoid process	Linear roughening on midshaft of humerus on medial side	Musculocutaneous nerve [C5, C6, C7]	Flexor of the arm at the glenohumeral joint
Biceps brachii	Long head—supraglenoid tubercle of scapula; short head—apex of coracoid process	Radial tuberosity	Musculocutaneous nerve [C5, C6]	Powerful flexor of the forearm at the elbow joint and supinator of the forearm; accessory flexor of the arm at the glenohumeral joint
Brachialis	Anterior aspect of humerus (medial and lateral surfaces) and adjacent intermuscular septae	Tuberosity of the ulna	Musculocutaneous nerve [C5, C6]; (small contribution by the radial nerve [C7] to lateral part of muscle)	Powerful flexor of the forearm at the elbow joint

The posterior compartment contains one muscle:

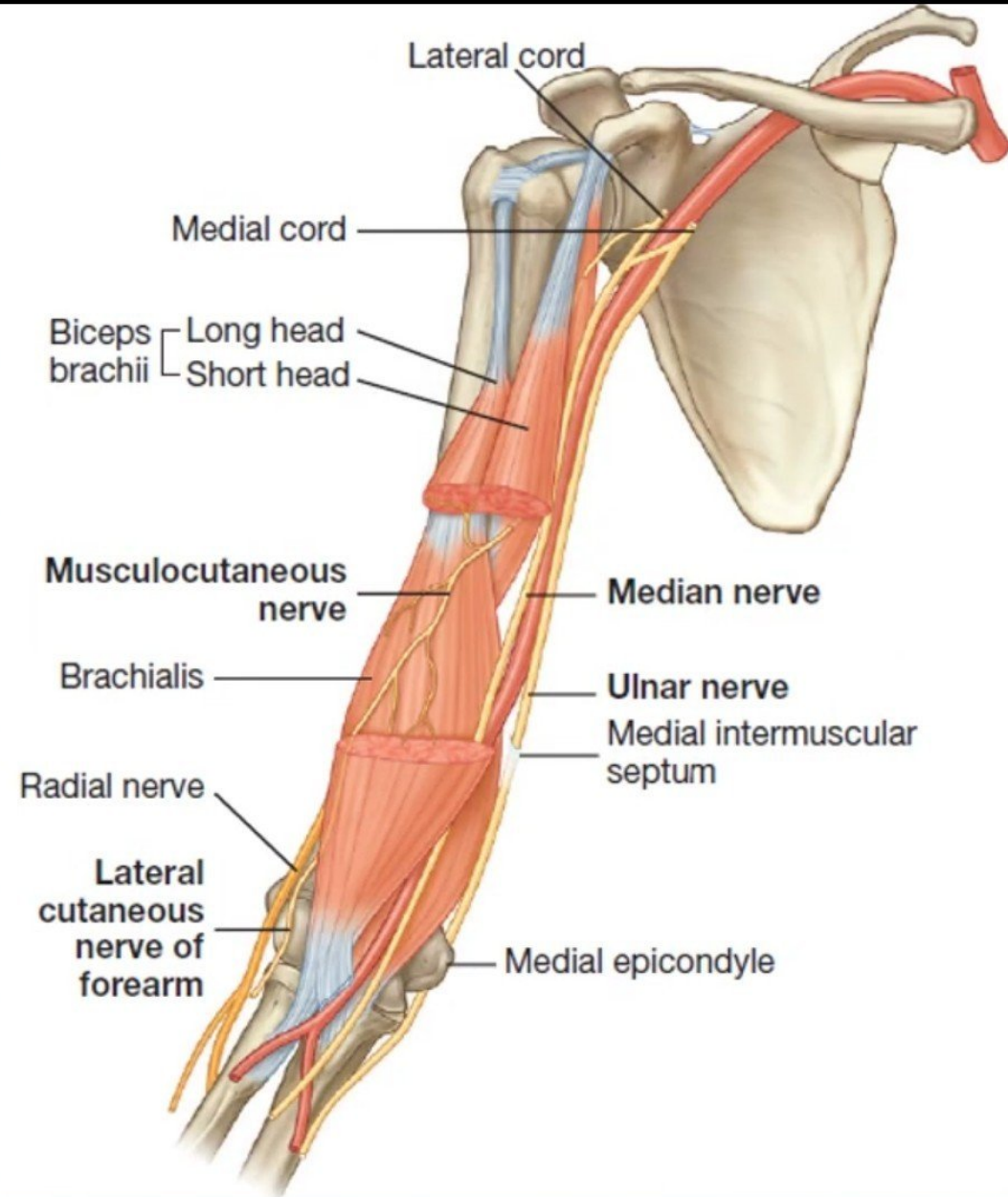
- **The triceps brachii** muscle
- **Origine:**
 - Long head : infraglenoid tubercle of scapula.
 - Medial head: posterior surface of humerus.
 - Lateral head : posterior surface of humerus.
- **Insertion:** Olecranon.
- **Function:** Extention of the forearm at the elbow joint: long head can also extend and adduct the arm at the shoulder joint.
- **It is innervated** by **the radial nerve (C6, C7, C8).**



Nerves

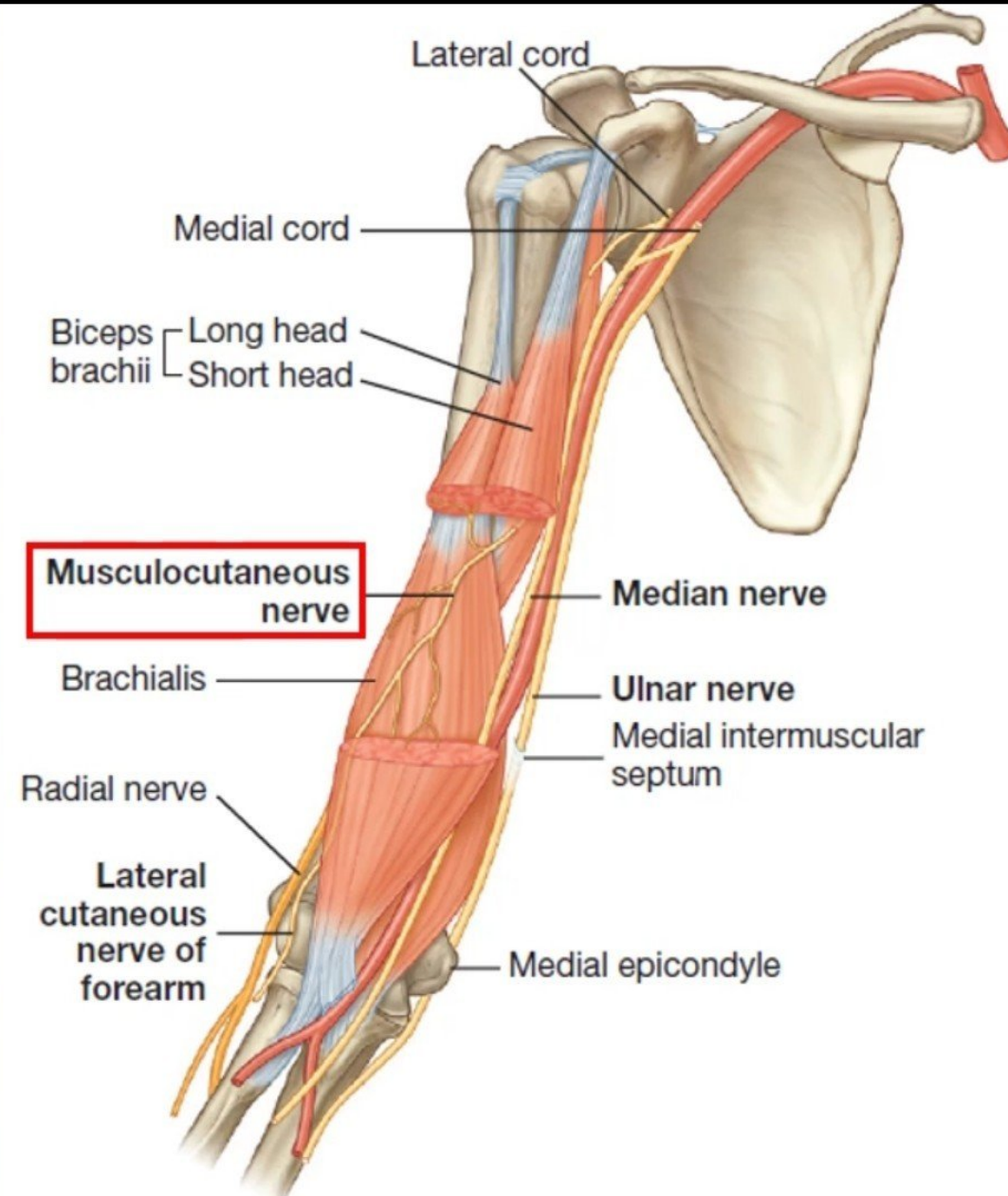
1. Musculocutaneous nerve
2. Median nerve
3. Ulnar nerve
4. Radial nerve

- The course & muscle innervated



1. Musculocutaneous nerve

- It pass through **the coracobrachialis muscle**.
- It passes diagonally down the arm in the plane **between the biceps brachii and brachialis muscles**.
- It continues as the **lateral cutaneous nerve of forearm**.
- Provides:
 - **Motor innervation** to **all muscles in the anterior compartment of the arm**.
 - **Sensory innervation** to skin on the **lateral** surface of **the forearm**.

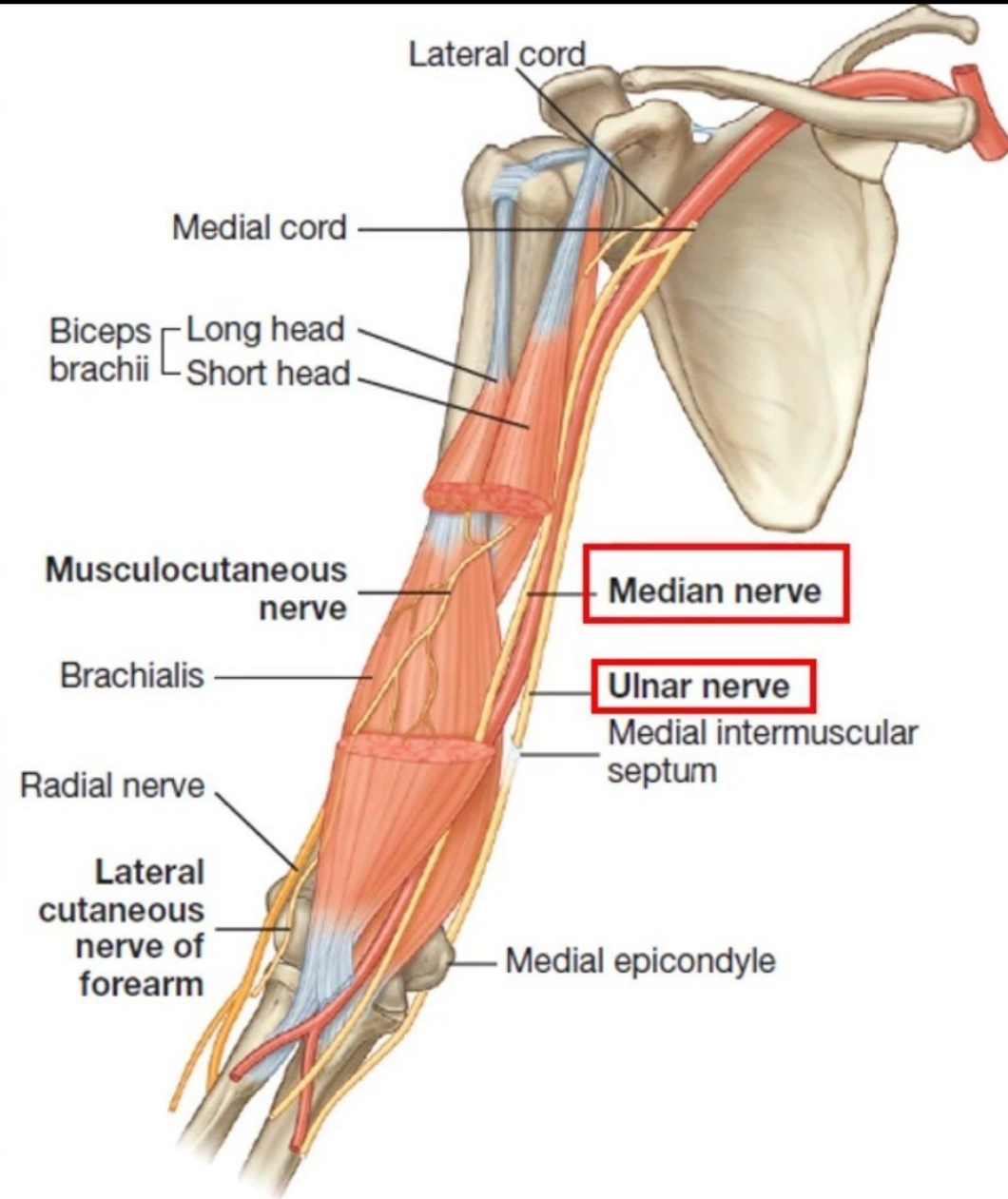


2. Median nerve

- Passes vertically down the **medial side of the arm** in the anterior compartment.
- It is related to **the brachial artery** throughout its course.
- It has **not major branches in the arm**.

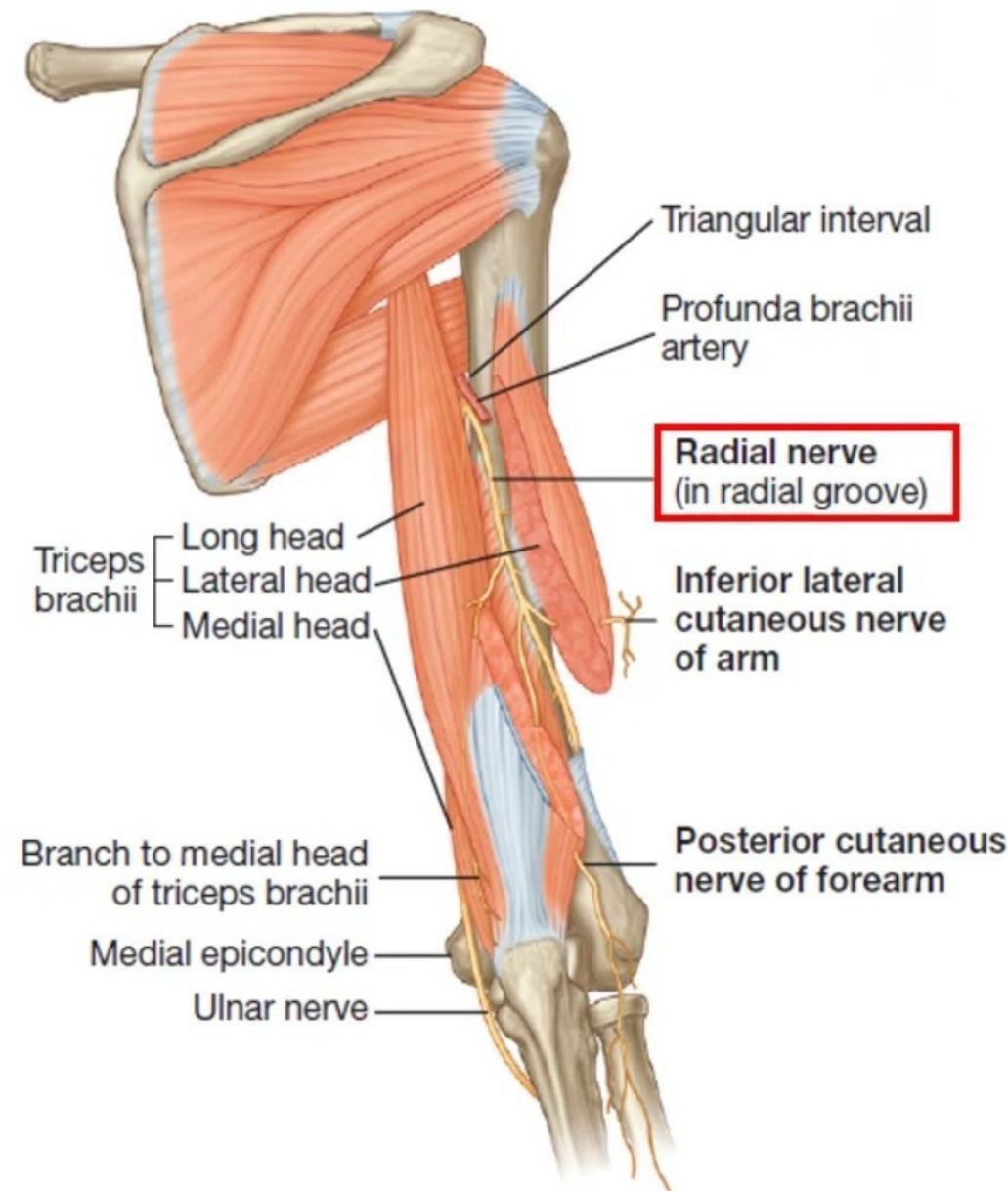
3. Ulnar nerve

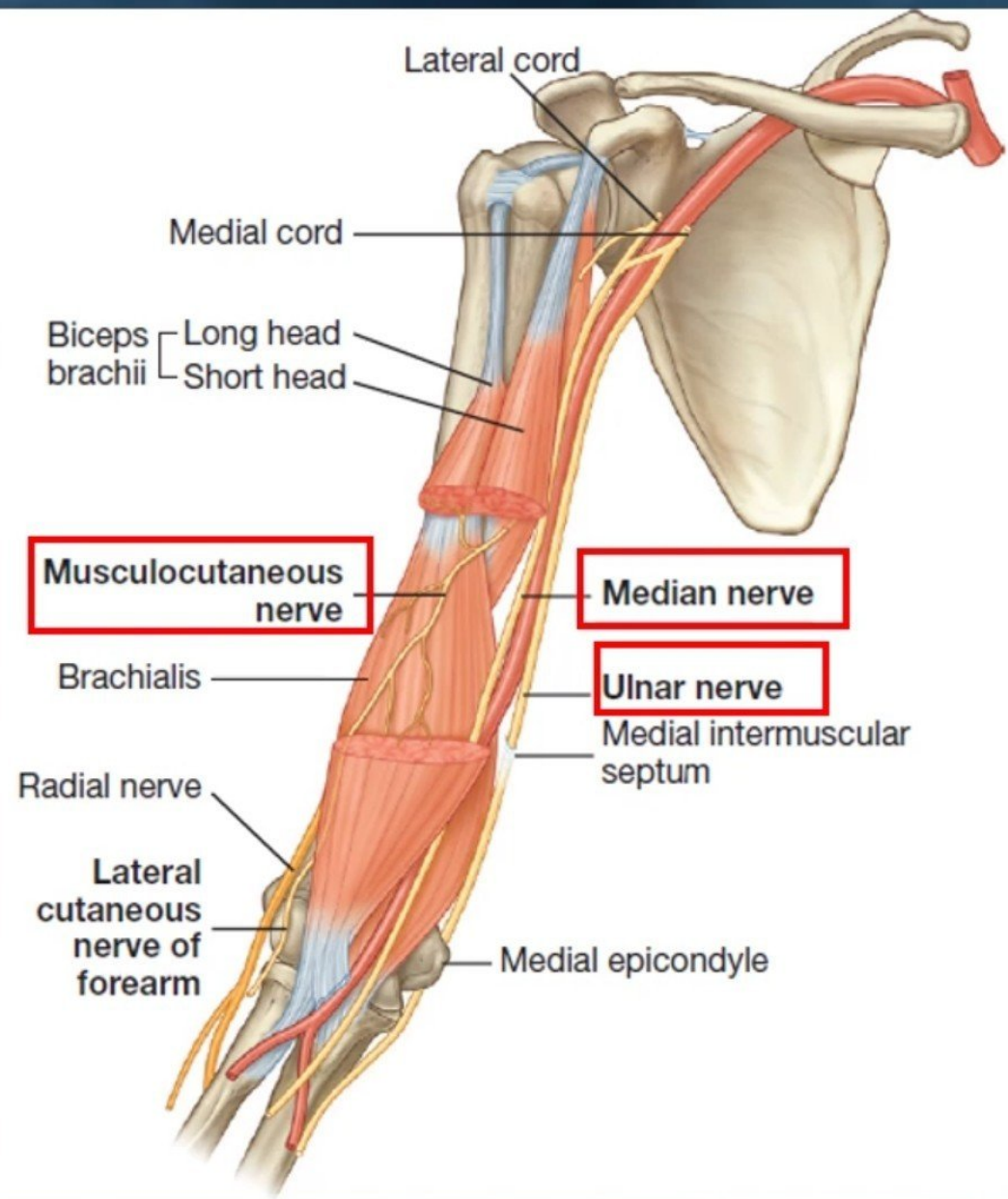
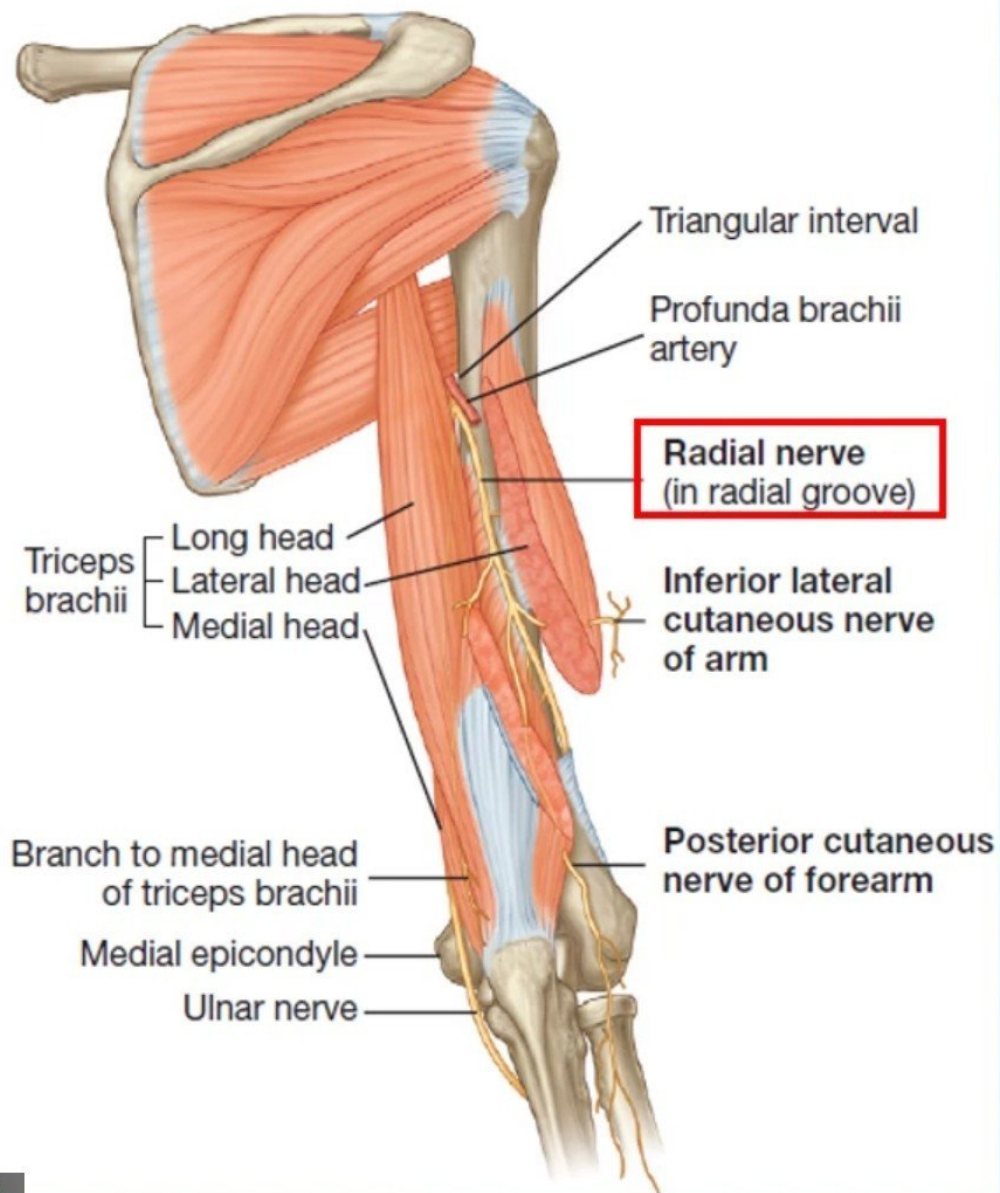
- It **penetrates the medial intermuscular septum** and **enters the posterior compartment**.
- It **passes posterior to the medial epicondyle** of the **humerus**.
- It has **no major branches in the arm**.



4. Radial nerve

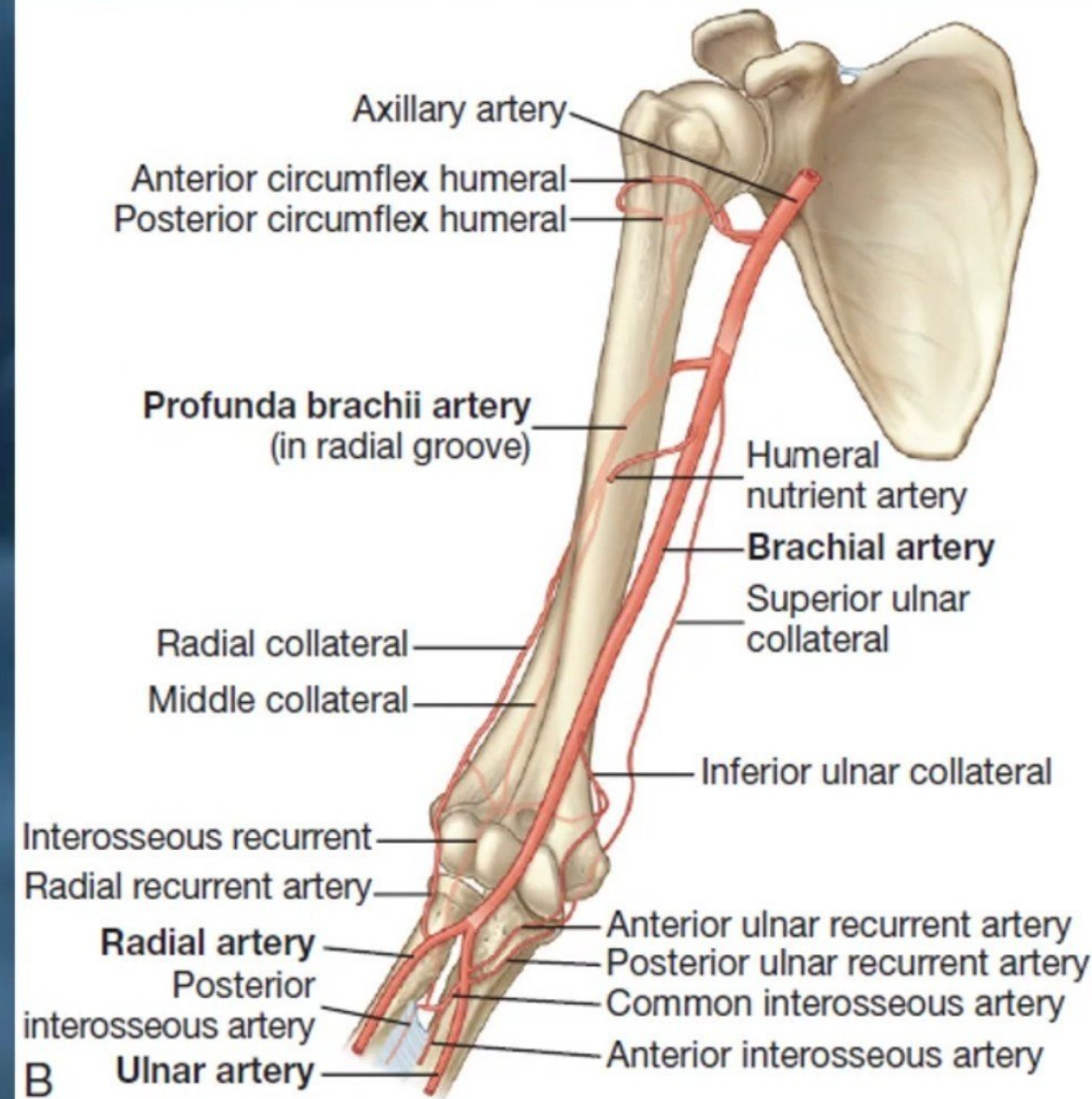
- It is accompanied by the **profunda brachii artery**.
- It enters the posterior compartment by passing through **the triangular interval**.
- It lies in **the radial groove** directly on bone.
- it **passes anteriorly** through the lateral intermuscular septum and enters the **anterior compartment** where it lies **between the brachialis and the brachioradialis** muscles.
- enters the forearm **anterior to the lateral epicondyle** of the humerus, just **deep to the brachioradialis muscle**.





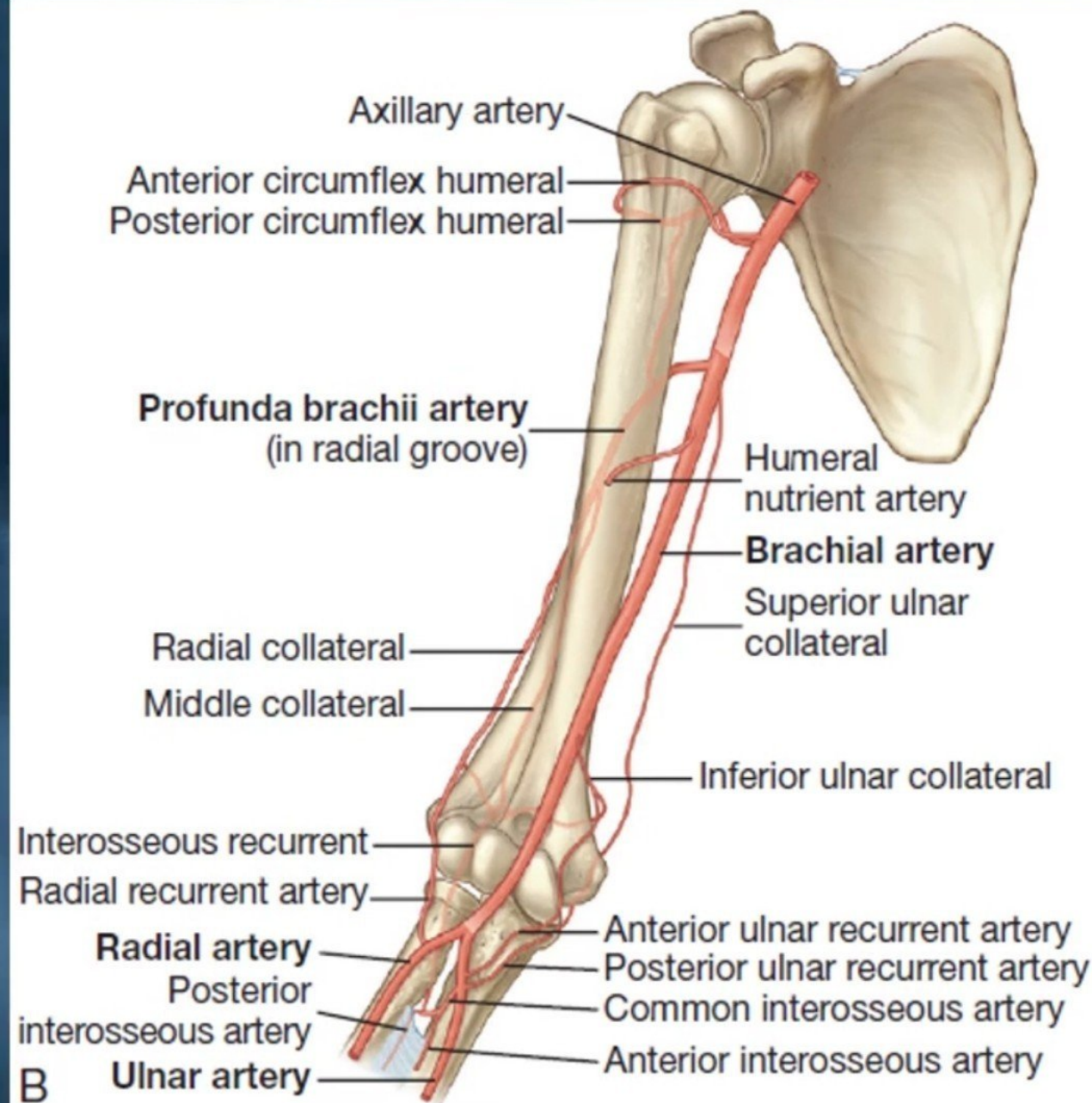
1. Brachial artery

- It is found in **the anterior compartment**.
- Beginning as a **continuation of the axillary artery** at the **lower border of the teres major muscle**.
- It terminates just distal to the elbow joint where it **divides into the radial and ulnar arteries**.
- In **the proximal arm**, the brachial artery lies on **the medial side**.
- In **the distal arm**, it crosses anteriorly to the elbow joint where it lies immediately medial to the tendon of the biceps brachii muscle.

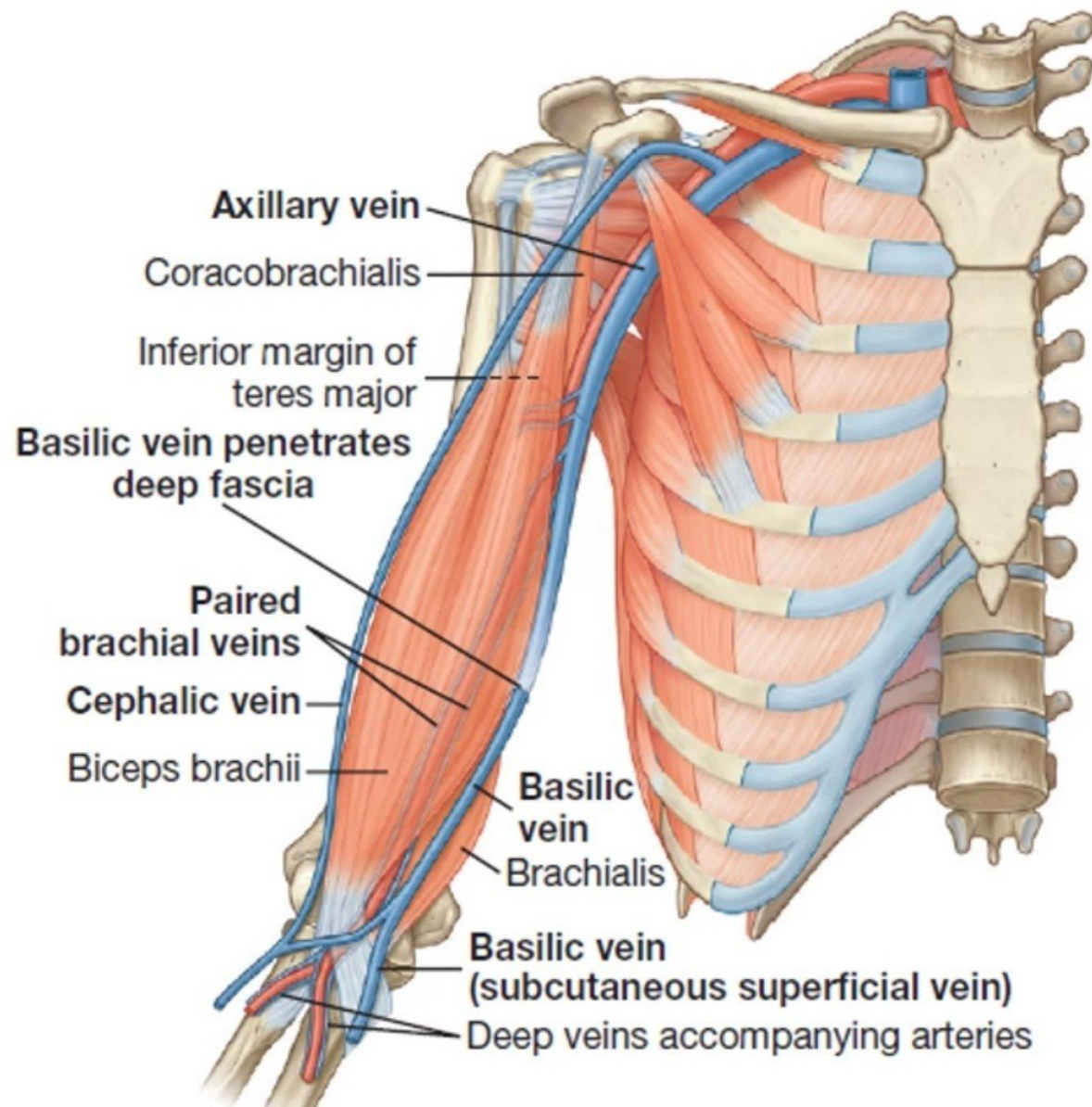


2. Profunda brachii artery

- It is **largest branch** of the brachial artery.
- It passes into and **supplies the posterior compartment of the arm.**
- It enters **the posterior compartment** with the **radial nerve** and together they pass through the triangular interval pass along the **radial groove.**

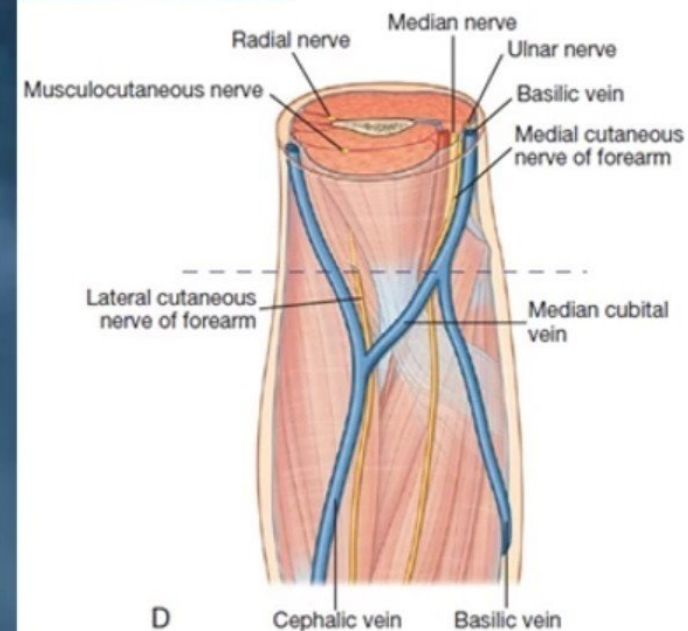
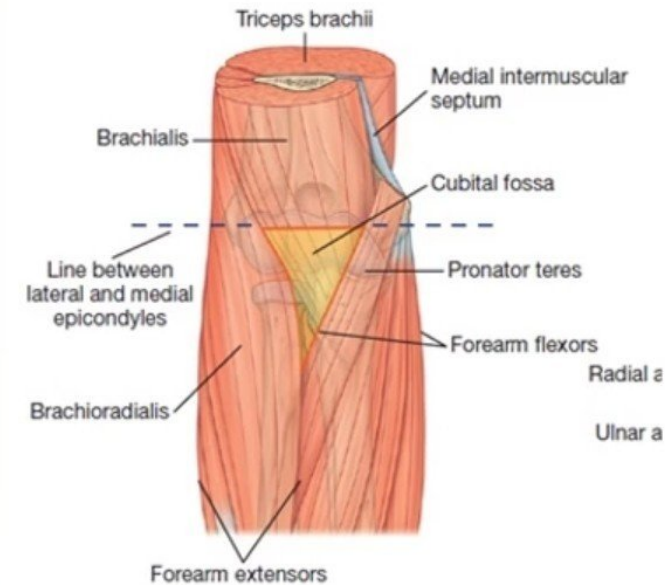


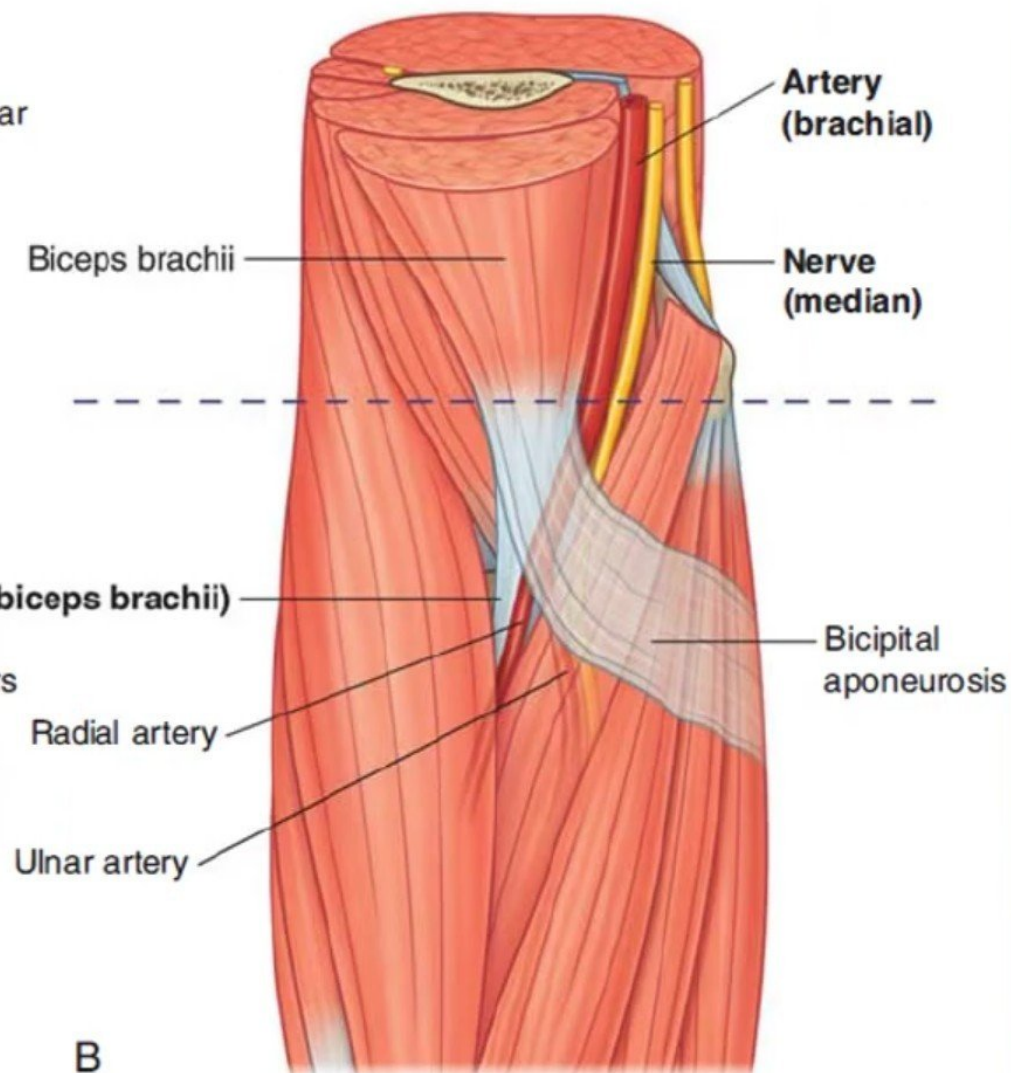
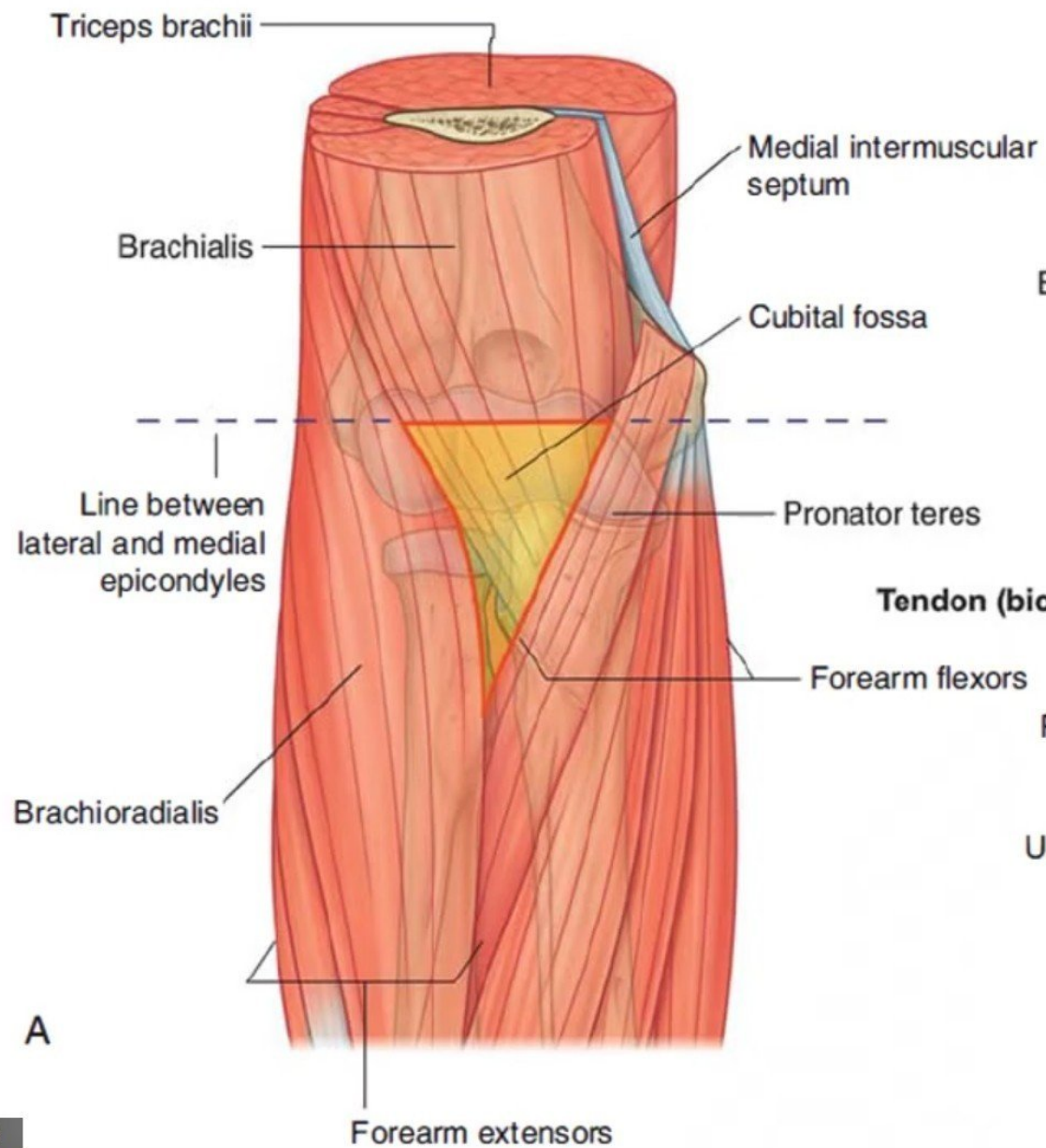
- **Paired brachial veins** pass along the **medial** and **lateral sides** of the brachial artery .
- **Two large subcutaneous veins:**
 1. **The basilic vein :** passes vertically in the distal half of the arm, penetrates deep fascia to assume a **position medial** to the brachial artery, and then becomes the **axillary vein** at **the lower border of the teres major muscle**. The brachial veins join the basilic, or axillary vein.
 2. **The cephalic vein :** passes superiorly on **the anterolateral aspect** of the arm and through the anterior wall of the axilla to reach **the axillary vein**.

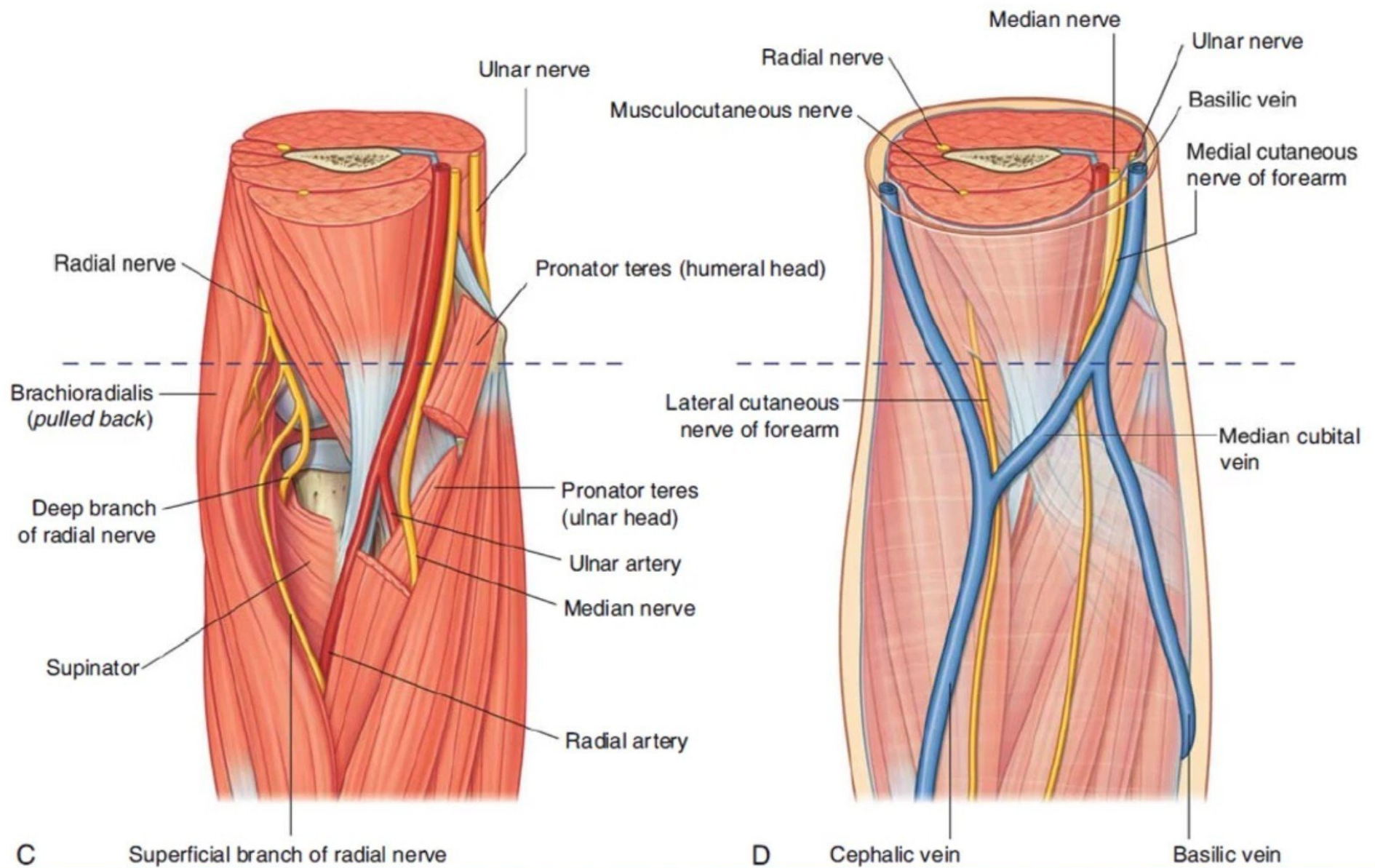


The cubital fossa

- It is an important area of **transition between the arm and the forearm**.
- It is a **triangular depression** that is located **anterior to the elbow joint**.
- **Boundaries:**
 - **The base:** is an imaginary horizontal line between the **medial and lateral epicondyles**.
 - **Laterally:** the **brachioradialis** muscle.
 - **Medially:** the **pronator teres** muscle.
- **The floor :** the **brachialis** muscle.
- **The roof :** the **bicipital aponeurosis** , the **median cubital vein** , **superficial fascia** and **skin**.







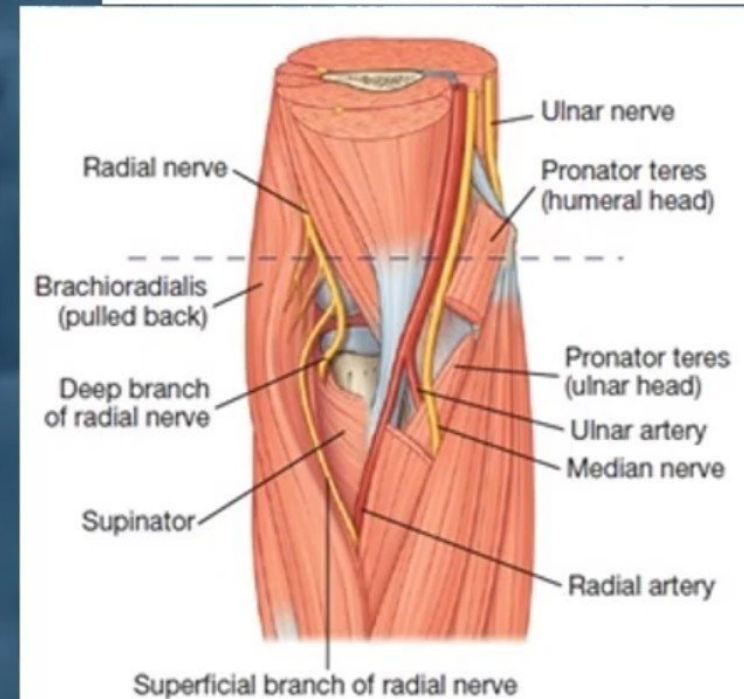
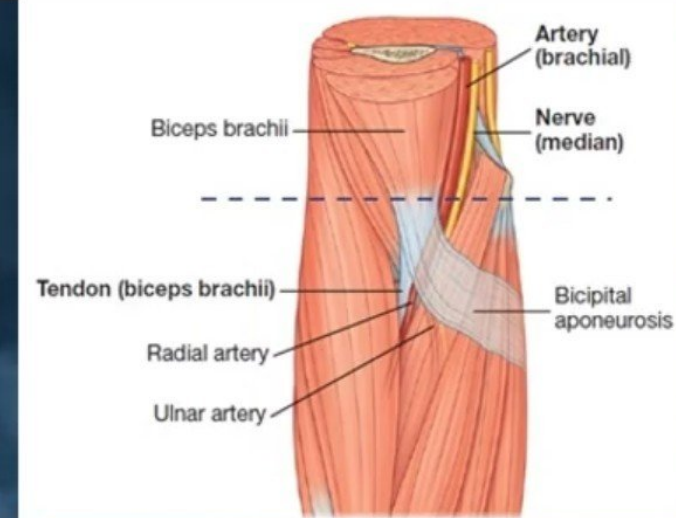
The contents of the cubital fossa

• The contents, from lateral to medial, are : (TAN)

1. The **tendon** of the biceps brachii muscle
2. The **brachial artery** (normally bifurcates into the radial and ulnar arteries in the apex of the fossa)
3. The **median nerve** (leaves the fossa by passing between the ulnar and humeral heads of the pronator teres muscle)

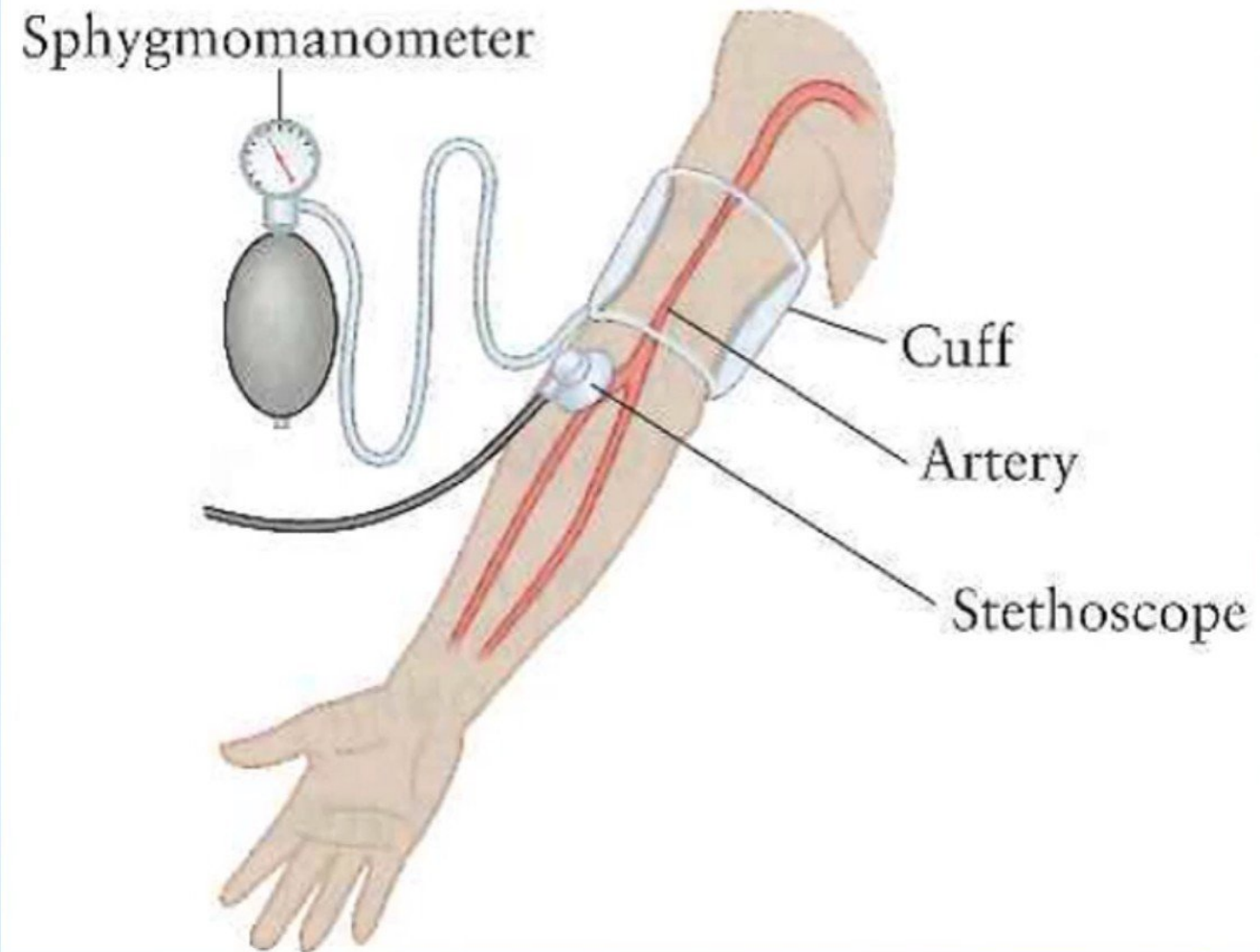
➤ The **radial nerve** lies just under **the brachioradialis muscle**.

➤ The **ulnar nerve** does not pass through the cubital fossa.



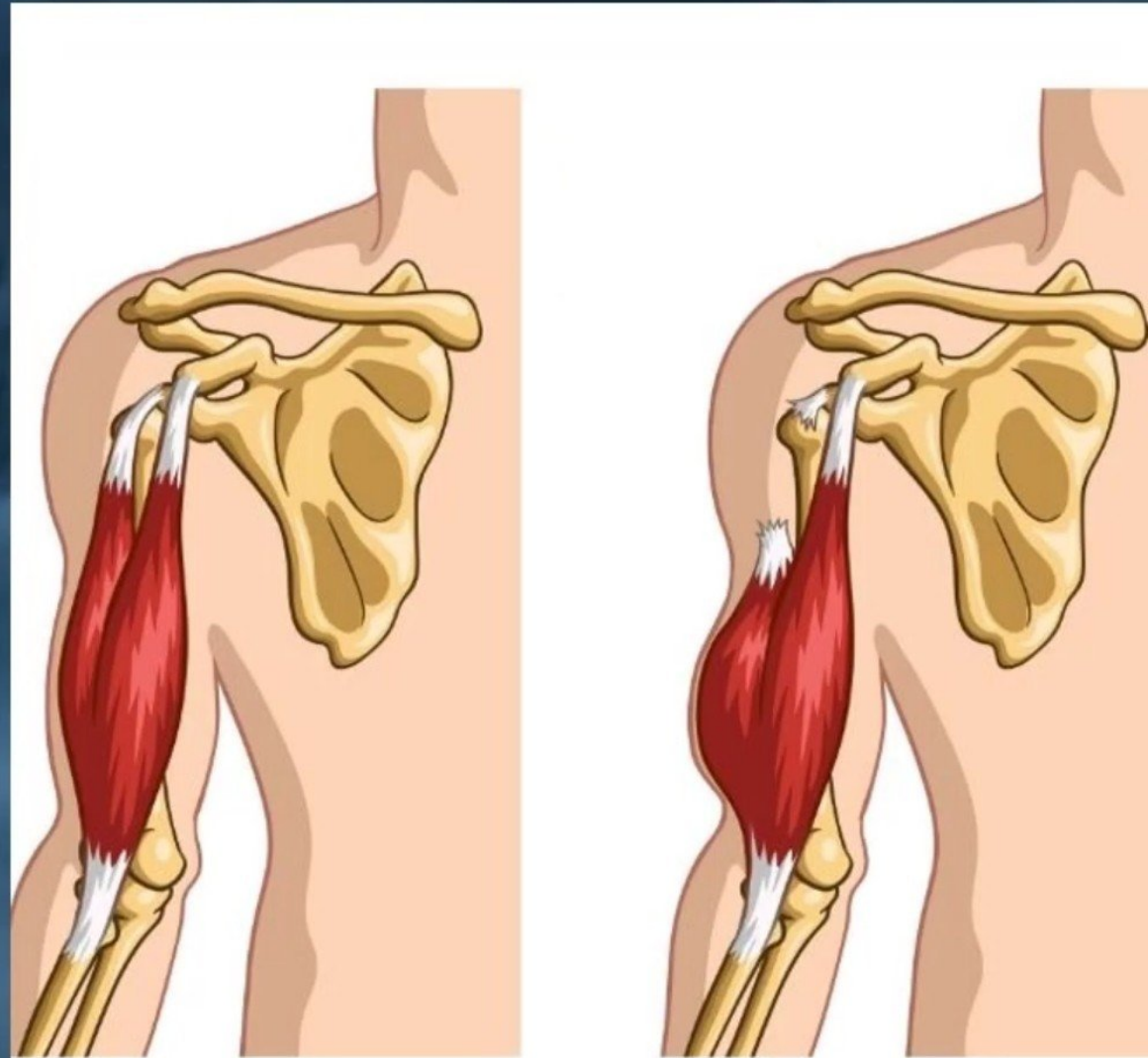
Auscultation of brachial artery pulsation

- When **taking a blood pressure reading** from a patient, the clinician places the **stethoscope over the brachial artery** in the **cubital fossa**.



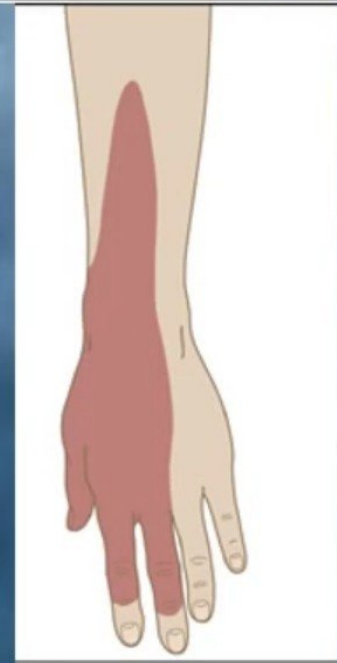
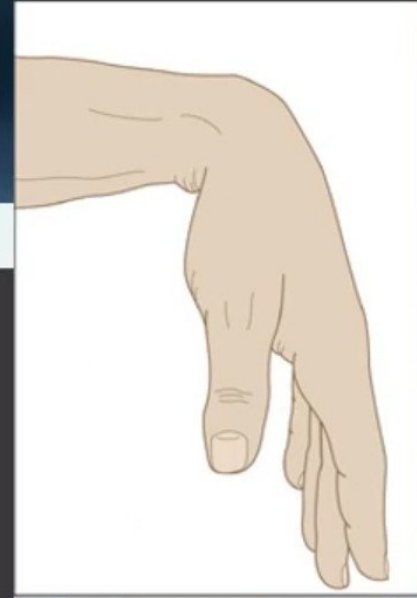
Rupture of biceps tendon

- most commonly involve is the tendon of the **long head** of the biceps brachii muscle.
- it produce a characteristic **deformity on flexing the elbow**, there is an extremely prominent **bulge of the muscle**— the “**Popeye**” sign.



Radial nerve injury in the arm

- It is tightly related with **the profunda brachii artery** between the medial and lateral heads of the triceps brachii muscle in the **radial groove**.
- a fracture of the **midshaft of the humerus**.
- **Nerve damage** and **loss of function**.
- The patient's symptoms usually include **wrist drop** (due to **denervation of the extensor muscles**) and sensory changes over **the dorsum of the hand**.





Thank you